Improving Customer Segmentation and Targeted Marketing Strategies through Data Analytics Techniques

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A report submitted as part of the requirements for Masters degree
in Information Technology With Business Intelligence
at the School of Computing
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Aberdeen, Scotland

August 2024

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Abstract

This work, with a dataset of existing customers, explores customer segmentation using two clustering algorithms: K-Means and DBSCAN. The dataset comprises demographic and spending score. K-Means was implemented using the Elbow method to determine the optimal number of clusters needed for the algorithm to run efficiently. It resulted in six (6) distinct groups in the dataset. However, K-means has a limitation which is its inability to handle noise and it came out with a silhouette score of 0.24. DBSCAN on the other hand identified 218 clusters and effectively managed the noise in the dataset. It isolated 2,530 data points out of a total row of 6,718. It gave a higher silhouette score of 0.317. A simple web application was developed to visualize and analyze clustering outcomes, providing insights for targeted marketing strategies. This is an attempt to give users who ordinarily wouldn't be able to run data analytics algorithm the opportunity to do it without the rigour of data analytics knowledge. Future enhancements to the the work could include refining customer labeling so as to be able to identify each group clearly and optimizing the application functionalities.

Acknowledgements

Firstly, I give thanks to God for the gift of health and strength that saw me through this journey.

My express gratitude to my supervisor Dr. Carlos Moreno-Garcia for his guiance and encouragement throughout this work. To my wonderful uncles; Adewale Asifat and Akanbi Luqman, the Hammeds, the Olowolas and my parent, your prayers did not go to waste. I say a big thank you to you all.

To my amazing cousins; Sijuade Halima, Bisola Akande-Taiwo and Adelabu Adewale, I can't express how much I appreciate you. Your unwavering belief in me was the push I needed to get through.

To friends who turned family; Alao Muyiwa, Ajoke Saliu, Babatunde Shuaib, Ajua Kyeremantin, Olutayo Ajibade and every one that I crossed path with in the course of the programme, I say thank you.

Declaration

I confirm that the work contained in this MSc project report has been composed solely by myself and has not been accepted in any previous application for a degree. All sources of information have been specifically acknowledged and all verbatim extracts are distinguished by quotation marks.

Signed .... Date ....

Akanbi Abdullahi Adesunkanmi

Contents

List of Tables

List of Figures

List of Algorithms

Listings

Chapter 1

Introduction

This chapter explains the projects goal. Overall, it shows the description of the report and gives a clear overview of the research and analysis done in the course of the work to ascertain its deliverables.

1.1 Background

According to Xixi and Chen, "ensuring product quality, providing high-class service and analyzing and predicting customer activities are the most important thing to the enterprise marketing" (2016 pp. 203). Businesses today realise the need to do more than put products out there and wait for customers to buy. There are lots of factors that come to play when you mention marketing, products, and customers. Businesses want to know what prompt their customers buying pattern. They want to know why and when their customers will buy certain product. They do this because they want to be shelve-ready to meet the demands of their customers. For instance, they want answers to questions like "What items do female customers buy?" or "What do consumers(married and 35-45 aged) prefer mostly?". They also want to know for instance if female customers purchase feta cheese with a percentage of 60% whereas male customers purchase tomato with a percentage of 46% (Gulluoglu 2015 p.154) . Hence, the need for customer segmentation and targeted marketing strategies. If businesses can effectively categorize customers based on their buying patterns and choices, they can direct their effort to individuals or certain group of customers and this may lead to more sales, more satisfied customers and ultimately, more profit. Retail marketers are constantly looking for ways to improve the effectiveness of their campaigns. One way to do this is to target customers with the particular offers most likely to attract them back to the store and to spend more time and money on their next visit. Demographic market segmentation is an approach to segmenting markets. (Gulluoglu 2015) Data analytics is a process where raw data is examined to get information so as to arrive at decision. This process involves applying several trained and tested algorithms to analyse, transpose and interpret dataset. With knowledge like this, organizations are able to make informed decisions, study trends of products and customer behaviours to reveal patterns that are not visible to the ordinary eyes. According to Aggarwal, Data mining is the study of collecting, cleaning, processing, analyzing, and gaining useful insights from data. A wide variation exists in terms of the problem domains, applications, formulations, and data representations that are encountered in real applications" (2015, pp. 1). It is safe to say that data mining is an umbrella term for any activities relating to data that seek to get insight, analyze, report and help manage informed business decisions. "So, like other companies, they know what products their customers want, but they also know what prices those customers will pay, how many items each will buy in a lifetime, and what triggers will make people buy more" (Davenport 2006 p. 2) The concept of segmenting customers has been around for long. You cannot address customers' need if you do not understand their behaviours and buying pattern. By segmentation, we are talking about classifying customers into groups. Each group will reflect the characteristics of different customers. "Customer segmentation refers to the existing customers being divided into different customer groups according to certain standards" (Xixi and Chen 2016 p.203) When customers are properly segmented, business owners can effectively design marketing advertisement that relates with each of the groups. This will ultimately lead to improved sales and higher profit margin. knowing that 60There are several data analytics techniques that are available to help decipher the insights hidden in customer data and can be explored to achieve segmentation. There are algorithms that use techniques such as clustering, association rule, and classification both supervised and unsupervised. SUPERVISED LEARNING (prediction) is a type of machine learning where the algorithm is trained on a labelled dataset. It learns to map input features to corresponding output labels. UNSUPERVISED (knowledge discovery) i.e., Finding patterns in unlabelled data (RGU 2024).

1.2 Aims

The major aim of this project is to improve customer segmentation and enhance targeted marketing campaign through the application of advanced data analytics techniques. This project seeks to create the leverage that will allows any business to gain a clear understanding of how customers behave and tailor their marketing strategy accordingly.

1.3 Objectives

1. To analyse and segment customers based on their buying patterns and behaviour using data mining and clustering algorithms. This will be achieved by gathering and processing a large customer dataset to find out the similarities in customer purchasing behaviours.

2. To evaluate the performance of selected data analytics tools used for customer segmentation. By this, 2 data analytics tools will be assessed to determine how effective they are in proper customer segmentation and the most suitable for the available dataset. 3. To build a simple user interface that allows users to interact with the dataset, perform analyses and visualize results without the need for extensive data processing and data analytics knowledge. This will be a verile tool in the hand of any business.

1.4 About this Thesis

This is the thesis of *Akanbi Abdullahi Adesunkanmi*, submitted as part of the requirements for the degree of MSc Information Technology With Business Intelligence at the School of Computing and Engineering, Robert Gordon University, Aberdeen, Scotland.

This thesis will investigate the details of customer segmentation and enhanced targeted marketing campaigns by exploring data analytics techniques. It will take a cursory look at some previously done works in the same subject of customer segmentation. It will also do a practical application of these techniques on a dataset. These datasets will be analysed using the selected analytical techniques to identify the customer segments.

1.5 Chapter List

Provide a list of all the chapters within the thesis and a brief summary of the content.

Chapter 2 Background Research. This chapter deals with existing literature on customer segmentation.

Chapter 3 Design. This chapter describes the dataset, processing, and cleaning.

Chapter 4 Implementation. This chapter deals with the analysis and the result.

Chapter 5 Evaluation & Testing. This chapter deals with the evaluation and testing of the findings in Chapter 4.

Chapter 6 Conclusion. The conclusions of the thesis are presented.

Chapter 2

Background Research

2.1 Review of Existing Literature

There has been a plethora of papers and work done in the field of data analytics and machine learning. This chapter seeks to review the previous works done, their findings and the different techniques available for this field of study. The chapter will look at the dataset available in the field, the collection process, the techniques available, evaluation and a look at the library available for a simple user interface that allows an administrative interaction for anyone who wishes to apply any of the algorithms to their dataset but does not have the requisite knowledge of data analytics.

2.2 Data Collection

An analysis is as good as its dataset. Data collection is an integral part of any data analytics work. The process involves gathering different information from various sources, making sure they are accurate and correct then proceeding to prepare the information for analysis. A well-sourced data is crucial in deriving insight from data thereby giving clear information. As clearly put by Aggarwal (2015), data collection methods include surveys, transaction records, social media integration, and sensor data, among others.

Data mining is a non-trivial process of extraction of information which is hidden, previously unknown and potentially useful, from large databases. Data mining can also be explained as finding the correlations in a large relational database based on the different depths of angles we analyze it. It is a powerful tool with high potential that helps organizations or companies increase their sales and gain more profit from the information about the dealings of their customers. (Aggarwal 2015)

Researchers in the past, while studying customer segmentation, have used different datasets. Xixi and Chen for example, used records of transaction from an e-commerce platform to study customer behaviour. The dataset used by Xixi and Chen covers information on product categories, customer demographics, product names and the frequency of purchase. “RFM model is one of the important means to measure customers’ value profitability. RFM model was first proposed by Arthur Hughes’s, which includes three important variables of measuring customers’ value of an e-commerce website: R recent consumption, F consumption frequency, M the amount of consumption” (Xixi and Chen 2015 p.203).

Gulluoglu on the other hand used a dataset collected from a retail store. The dataset covers customer information, their transactions and loyalty programme information. (Gulluoglu 2015). These datasets afforded the researchers a buoyant source of information for studying and learning the customers’ buying preferences and patterns.

Generally, datasets used in customer segmentation work cover demographics, purchase history and buying behaviour. In the work of Xixi and Chen, customer transaction records were analysed and they presented insights into buying patterns. The demographic data reveals the peculiarities of each customer group. When handling these datasets, it is important to prepare them in a way that will be usable. Some datasets may have missing values or inconsistent data types that will make calculations difficult. Hence the need for data pre-processing. While explaining the importance of data pre-processing, Kotu and Deshpande (2014) both mentioned that it is crucial to preprocess the collected data to handle missing values, inconsistencies, and noise. This pre-processing step will ensure dataset is reliable for analysis.

Before algorithms are used, there is often a certain amount of pre-processing of the data that is required. These pre-processing include dealing with missing values, attribute selection, one-hot encoding, and dealing with different scales (numeric values that require normalisation, centre, scale and standardisation) (RGU 2024). “The purpose of pre-processing is to transform the raw input data into an appropriate format for subsequent analysis” (Tan et al. 2018 p. 25). They further identified several steps involved in the data pre-processing. Some of them are merging different data from multiple sources into one large dataset, cleaning the data so as to get rid of duplicate values and selecting data that are relevant to the task at hand. Oftentimes, we do not need all the information in a dataset for the task at hand so there is a need for removal of certain data to make the dataset fit into the right use.

2.3 Data Analysis Techniques

In order to get useful insight from raw data, there are several analysis techniques. Among these techniques are machine learning algorithms and statistical analysis. Clustering, Classification and Association rules are very effective mining techniques for customer segmentation. (Tan et al. 2018) According Tan et al. (2018 p.24) “Not all information discovery tasks are considered to be data mining. Examples include queries, e.g., looking up individual records in a database or finding web pages that contain a particular set of keywords” Data mining is the process of automatically discovering useful information in large data Repositories, he further asserts.

Data mining is the process of extracting information, patterns, and insights from large datasets using various techniques and algorithms. Data mining requires a large dataset most times, a machine learning algorithm and an evaluation. (RGU 2024).

Customers can be grouped based on characteristics that are similar using the clustering technique. K-means and DBSCAN are among the common algorithms used for this purpose. According to XU and WUNSCH “The step is usually combined with the selection of a corresponding proximity measure and the construction of a criterion function” (2005 p. 645). This clearly explains the process called clustering. The algorithm makes a choice of an appropriate proximity metric and it constructs a function as to how it wants the data points to be selected. Clustering, which is often referred to as unsupervised classification, the data available are usually not labelled. “The goal of clustering is to separate a finite unlabeled data set into a finite and discrete set of “natural,” hidden data structures, rather than provide an accurate characterization of unobserved samples generated from the same probability distribution” (XU and WUNSCH 2005 p. 645)

it is clear that there is no universally agreed-upon number of clusters that an algorithm has to generate but one thing cuts across all the scholars, it is the agreement that when an algorithm is run against a dataset, it arbitrarily partitions them into several clusters. XU and WUNSCH (2005) suggest that clustering dates back to the days of Aristotle and that it organizes data into a particular structure according to proximity to other data in the given dataset.

When we predict the category to which certain observations belong based on a trained dataset, that involves classification. Hastie, Tibshirani and Friedman (2009) assert that decision trees, random forests and logic regression are widely used algorithms in these classifications. Association rule mining helps in establishing the relationships between variables that exist in a dataset. This is useful for market basket analysis as seen in Agrwala and Srikant (1994) where the goal is to identify items that co-exist in different independent transactions.

Xixi and Chen (2016) segmented customers based on their purchasing behaviour using the K-means clustering algorithm. They identified distinct customer segments, such as frequent buyers, one-time buyers and occasional buyers. The algorithm grouped the customers in the dataset similar buying patterns. This allows the business to tailor marketing strategies to each group's preferences.

Demographics and purchasing behaviour played a vital role in classification model. Classification models predict category which new observation belongs based on a dataset that has been trained. Gulluoglu (2015) used decision trees and random forests to classify customers using demographic and purchasing pattern data. The model is used to identify customers that are high valued, and it can predict how they will behave in the future with certain product. He used metrics such as accuracy, precision, recall and the F1 score to evaluate the performance of the classification.

Ester et als' demostration show that DBSCAN is very effective when using dataset where clusters are of varying shapes and densities. One clear advantage is has is the ability to detect noise points in a dataset (Ester et al. 1996). They tried to compare DBSCAN with CLARANs in terms of effectivity and they did this using three synthetic sample databases. "DBSCAN discovers all clusters and detects the noise points according to definition from all sample databases. CLARANS,however, splits clusters if they are relatively large or if they are close to some other cluster. Furthermore, CLARANS has no explicit notion of noise. Instead, all points are assigned to their closest medoid." (Ester et al. 1996 p.230)

2.4 Evaluation

Data mining techniques must be rigorously evaluated to ensure their effectiveness and reliability in real-world applications (RGU 2024). Data analytics process cannot be said to be complete without evaluating the effectiveness of the analysis models. Evaluation is crucial step in data analytics process. Evaluation metrics are important because they ensure that the model used perform well and infer decision from new data. Powers (2011) identified few metrics; accuracy, precision, recall and the F1 score as some of the common evaluation metrics for classification models. Metrics such as the Silhouette score, Davies-Bouldin Index, and Within-Cluster sum of squares are used to assess performance in clustering models (Rousseeuw, 1987).

Researchers use Cross-validation technique to validate the robustness of models. This is achieved by partitioning the dataset into subsets. A subset is used for training the models and the other subset for testing. To ensure the models performance is consistent, the process is repeated multiple times (Kohavi, 1995). When models are evaluated, the goal is not just to get the best of the models but it also allows for clear understanding of its potential biases and limitations. All models seem to have both if clearly evaluated.

Xixi and Chen (2016) used the silhouette score to evaluate their clustering model. The silhouette score measures how similar an object is to its own cluster in comparison to other clusters. A better-defined cluster will be determined by a higher silhouette score. They also used the within-cluster sum of squares to check how compact the clusters are. Clusters with lower values indicate more compactness and distinctiveness.

It is important to look at some of the works reviewed. Lets take a structured approach to consider their contribution to the field of customer segmentation. This aspect will focus on the differences in the algorithms used, their research mythologies and result obtained.

XU and WUNSCH (2005) focused on K-Means and hierarchical clustering in their work. In the book titled "Clustering" , they presented a comprehensive study that details these algorithms, showing their performance and use cases. They emphasize the theory behind clustering algorithms. They also exhibitited the different proximity measure and other functions affect clustering. The practical guidelines for selecting clustering methods is detailed by the authors. An overview of their work shows that K-Means is a better algorithm for datasets that are large. MacQueen, who is credited with the development of the K-Means algorithm focused on partitioning data into what he called the K clusters. This he said can be achieved by minimizing the Within-cluster variance. "the k-means procedure is easily programmed and is computationally economical, so that it is feasible to process very large samples on a digital computer" (MacQueen 1967 p. 281). It is clear how much has been spoken about K-means to be able to manage a large dataset and it is said to be computationally economical. He further discusses (MacQueen 1967) that k-means as an algorithm was originally devised in an attempt to find a feasible method of computing optimal partition. His work gave a clear understanding of the purpose of k-means. His point of view is not find a definitive grouping that is unique, rather an attempt to aid the user in finding qualitative and quantitative understanding of large amount of N-dimensional data by providing good similarity groups (MacQueen 1967). While Xixi and Chen, MacQueen, Xu and Wunsh focus on K-means, Agrwala and Srikant were more interested in the Association mining rule. Hastie, Tibshirani and Friedman are of the opinion that decision trees, random forests and logic regression are widely used for classifcation. Gulluoglu employed decision trees and random forests while classifying customers in a dataset with demographic and purchasing pattern.

Ester et al. instead decided to introduce the method which was named Density-Based Clustering of Application with Noise (DBSCAN).

Where K-means lags which is the noise detection area is properly accounted for in DBSCAN as it excels in noise detection. Also, K-means always requires the user to specify the number of optimum clusters after this must have been determined by the Elbow method, whereas DBSCAN do not need a predefined number of clusters but uses a two parameters; the Eps and MinPts.

In terms of efficiency and scalability, K-means has proven to be computationally efficient and scalable which gives it edge when it comes to datasets that are very large. DBSCAN can be computationally intensive for dataset that are of high dimentionality, though it is strong in noise handling.

2.5 Graphical User Interface for User Interaction

Graphical User Interface (GUI) allows users to do what they wouldn't have been able to do without the knowledge of stastical analysis and programming language. A simple and well-designed GUI boost the usability of data analytics application. In cases of customer segmentation, for example, a GUI allows users to interact seamlessly with the dataset, perform analyses and even visualize results without knowing the intricacies of writing of knowing any programming language. The R programming language is shipped with a package called Shiny. This package provides powerful tools for developing interactive GUIs.

Xixi and Chen (2016) developed an application using Shiny to visualize customer segments and their purchasing behaviour. The application gives users the room to filter data based on their age, gender and purchase frequency. Marketers can make data-driven decisions as the application allows for interactive plots and tables which provided a comprehensive view of customer segments.

Gulluoglu (2015) also used the R Programming shiny package to design a dashboard for customer segmentation analysis. The dashboard has built in features that supply real-time updates, interactive visualizations and reports. The application allows users to filter down into specific customer segments, view detailed profiles and analyse purchasing patterns.

A well-designed GUI should be intuitive, responsive and capable of handling large datasets efficiently (Pavia 2020). The development of any GUI in R programming language for customer segmentation takes lots of steps. These include designing the interface with users at the canter of the decisions made, integrating functions that process data and creating visualization. According to Pavia(2020 p. 2) while quoting Murrel (2019) “it is possible to work with R for years and remain unaware of important and useful features”. The R programming language is very powerful in doing some data analytics job and it is shipped with lots of packages that allow use to interact with the programme using its User Interface, Rstudio. Rstudio allows more flexibility and robustness.

The Shiny Packages is a web application framework for R. it makes it incredibly easy to build interactive web application with the R programming language. It comes with prebuilt widget that makes it possible to build beautiful interfaces with little effort.

2.6 Conclusions

This chapter has highlighted the few algorithms available to a data analyst who wants to do some classifications on a customer dataset. We have focused more on two among the common clustering algorithms. These are K-Means and DBCAN. The chapter has also displayed the strengths and the limitations of K-Means and DBSCAN which gives a clear picture for anyone who wishes to do a clustering exercise. The chapter will form a clear background for the other chapters in the work.

Chapter 3

Design

This chapter covers the methodology used in this work. It explains the dataset used, the data analytics technique employed and the process of transforming the dataset into a usable format. The K-means algorithm is used in this work to explain customer segmentation.

3.1 Dataset Description

For this work, we will be using a publicly available dataset from Kaggle.com. Kaggle is a free online platform for data and machine learning professionals. If provides a wealth of resources for users to learn and collaborate. The dataset was produced by an automobile company that wishes to launch a new product into the market but certainly would like to know the behaviour of its existing customers. By segmenting the customers, the company can make targeted marketing strategy that will focus on a desirable group. This will intern save them lots of money. The dataset is licensed to be publicly used. It gives users the right to copy, modify, distribute and perform the work, even for commercial purposes, all without asking permission. The licence can be found here: https://creativecommons.org/publicdomain/zero/1.0/ The dataset covers various customer purchasing behaviours and particular pattern. This provides a wealth of information for customer segmentation. The dataset contains the following attributes: Columns:

1. Unique ID: This is a unique identifier for each of the customers in the dataset.

2. Gender: Gender of the customer

3. Ever_Married: This is the marital status of the customers

4. Age: The age of the customers
5. Graduated: This shows weather a customer is a graduate or not
6. Profession: profession of the customer
7. Work_Experience: This tells the number of years the customer has been working
8. Spending Score: This is assigned to customer based on their purchasing behaviour
9. Family Size: The size of the family each of the customers belongs.

ID	Gender	Ever_Married	Age	Graduated	Profession	Work_Experience	Spending_Score	Family_Size
458989	Female	Yes	36	Yes	Engineer	0	Low	1
458994	Male	Yes	37	Yes	Healthcare	8	Average	4
458996	Female	Yes	69	No		0	Low	1
459000	Male	Yes	59	No	Executive	11	High	2
459001	Female	No	19	No	Marketing		Low	4
459003	Male	Yes	47	Yes	Doctor	0	High	5
459005	Male	Yes	61	Yes	Doctor	5	Low	3
459008	Female	Yes	47	Yes	Artist	1	Average	3
459013	Male	Yes	50	Yes	Artist	2	Average	4
459014	Male	No	19	No	Healthcare	0	Low	4
459015	Male	No	22	No	Healthcare	0	Low	3
459016	Female	No	22	No	Healthcare	0	Low	6
459024	Male	Yes	50	Yes	Artist	1	Average	5
459026	Male	No	27	No	Healthcare	8	Low	3
459032	Male	No	18	No	Doctor	0	Low	3
459033	Female	Yes	61	Yes	Artist	0	Low	1
459036	Female	Yes	20	Yes	Lawyer	1	Average	3
459039	Male	Yes	45	Yes	Artist	1	Average	2
459041	Male	Yes	55	Yes	Artist	8	Low	1
459045	Female	Yes	88	Yes	Lawyer	1	Average	4
459056	Male	Yes	63	No	Executive		High	3
459057	Male	Yes	69	No	Lawyer		High	
459058	Male	No	42	Yes	Artist	0	Low	4
459059	Male	Yes	79	No	Executive		High	2
459061	Female	Yes	35	Yes	Healthcare	9	High	3
459064	Male	Yes	27	No	Executive	5	High	4
459065	Male	Yes	52	Yes	Engineer		Low	2
459074	Female	No	29	Yes	Healthcare	0	Low	4
459077	Female	Yes	79	No	Lawyer	1	High	2
459079	Male	Yes	87	Yes	Lawyer		High	2
459080	Male	Yes	89	No	Lawyer	1	Low	2

Figure 3.1: Sample Dataset

The dataset is available on Kaggle with a licence listed under the Creative Commons Zero (CC0) Licence. This signifies that anyone can copy, modify, distribute and perform the work, even for commercial purposes, all without asking permission.

3.2 Data Processing

To ensure the data is suitable for analysis, there is need for the dataset to be processed and prepared. The following processes are implemented.

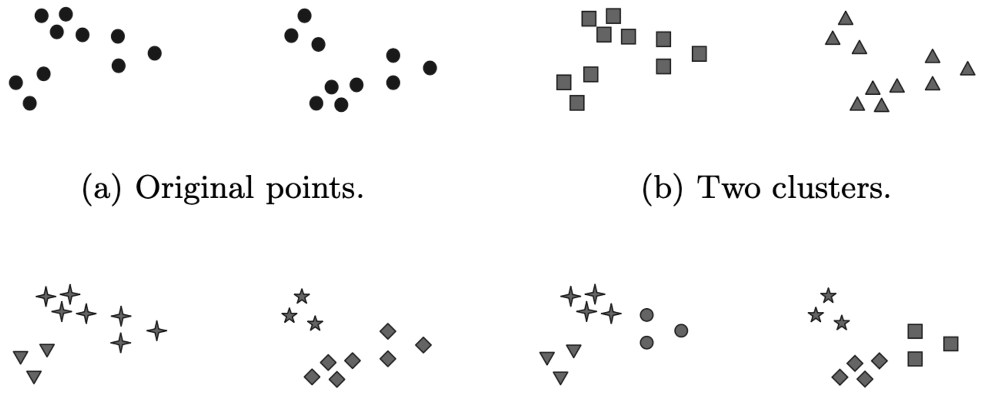
Missing Values: Missing Values can mar a supposedly good analysis. An analyst must cater for this before proceeding with their analysis. Missing values were identified and handled accordingly. The rows that have values integral to the analysis missing were removed. For most of the values, the missing values were replaced with "NA".

Transformation: this involves creating new features and modify existing ones to improve the structure of dataset. Normalization: this ensures that numerical features can contribute immensely to a clustering process and this is done by scaling variables to a standard range. K-means, which is one of the algorithm used in this work, generates clusters which are sensitive to the magnitude of data points. For the sake of this work, we have employed the Min-Max Scaling. This technique is used to scale character features, such as "work experience", "spending score" to a range of 0 to 1

Categorical Encoding: variables like "gender", "ever married", "graduated" and "profession" were encoded and converted into a usable format for the algorithms. They were scaled too like other variables.

3.3 Methodology

The method employed for this work is the clustering algorithm. "Clustering aims to find useful groups of objects (clusters), where usefulness is defined by the goals of the data analysis. Not surprisingly, several different notions of a cluster prove useful in practice. " (Tan et al. 2019 p. 313) An entire collection of clusters is commonly referred to as a clustering



(a) Original points. (b) Two clusters.

(c) Four clusters. (d) Six clusters.

Figure 3.2: Tan et al. 2019 Fig 5.1 Three different ways of clustering the same set of points

In this work, we have used two of the popular clustering techniques to explain the concept of customer segmentation. These two are K-means and DBSCAN

3.3.1 K-Means Clustering

K-Means is one of the most prominent prototype-based clustering techniques (Tan et al. 2019). Using the K-means clustering technique, it is important to first choose the K initial centroids, where K is a user-specified parameter. The user of the algorithm will be one to decide the amount cluster needed for the job. The algorithm then assign each of the K points selected to the closest centroid. Each of the collection of points assigned to a centroid is regarded as a cluster. The algorithm continues to update the clusters based on the centroid position and the points of the collection. This is repeated until no change in then clusters and the centroids remain the same (Tan et al. 2019). K-means is a clustering algorithm that partitions the supplied dataset into clusters based on centroid. Each of the data point belongs to the cluster with the nearest mean. The process follows a particular pattern; first it randomly selects the initial centroid, names it K. it then assigns each of the data point to the nearest centroid. Next step is to recalculate the centroids based on the mean of all points assigned to each of the cluster. This step is then repeated and it update the centroid and the data point till the best cluster is achieved. In K-Means algorithm, 5 steps are identified according to Tan et al. (2019); 1. Select K points as initial centroids. 2. Repeat 3. Form K clusters by assigning each points to its closest centroid 4. Recompute the centroid of each cluster

5. Until centroids do not change

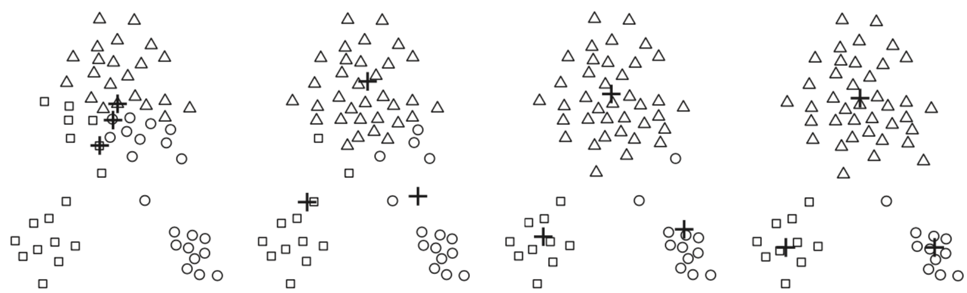


Figure 3.3: Tan et al. 2019 Fig 5.2 Clustering Stages

Steps 2, 3 and 4 are demonstrated in fig 2 (b), (c) and (d) respectively. At iteration 4, the algorithm terminates as there is no more change to the clusters. Points Assignments it is important that we understand how the algorithm assigns these data points to each centroids. “we need a proximity measure that quantifies the notion of “closest” for the specific data under consideration. Euclidean (L2) distance is often used for data points in Euclidean space, while cosine similarity is more appropriate for document” (Tan et al. 2019 p. 318) According to Aggarwal (2015 p. 162), “In the k-means algorithm, the sum of the squares of the Euclidean distances of data points to their closest representatives is used to quantify the objective function of the clustering”. Therefore, we have:

$$\text{Dist}(\overline{X}_i, \overline{Y}_j) = \|\overline{X}_i - \overline{Y}_j\|_2^2.$$

Aggarwal 2015 (6.5)

Here, $\|\cdot\|_p$ represents the Lp-norm. The expression Dist(Xi, Yj) can be viewed as the squared error of approximating a data point with its closest representative. Thus, the over- all objective minimizes the sum of square errors over different data points. (p. 186) To further clarify the formulae, it calculates the distance between two data points in clustering. The two points being Xi and Yj in K-means clustering. The distance is calculated using the squared Euclidean distance. The purpose here is to reduce as much as possible the sum of squared errors (SSE). This means we intend to find the best way to represent each of the data points by a “representative” point in its cluster. K-means is different from other clustering algorithm because the optimal representative is the mean of the points in the cluster.

Determining the Optimal Number of Clusters It is very important to be able to determine the appropriate number of clusters for K-means algorithm. This is crucial to to K-means clustering. There are there popular methods for namely; Elbow, Silhouette and Gap statistics. For the purpose of this work, we will be focusing on the Elbow and the Silhouette method. This methods involve running K-means clustering on the dataset for a range of K values and then proceed to plot the sum of squared errors (SSE) against the number of clusters. According to Bholowalia and Kumar (2014 p.18), "the Elbow method is a method which looks at the percentage of variance explained as a function of the number of clusters".

"This method exists upon the idea that one should choose a number of clusters so that adding another cluster doesn't give much better modelling of the data. The percentage of variance explained by the clusters is plotted against the number clusters." (Bholowalia and Kumar 2014 p.18) They identified 7 steps in the elbow method

1.Initialize $k=1$ 2. Start 3. Increment the value of $k 4. Measure the cost of the optimal quality solution 5. If at some point the cost of the solution drops dramatically 6. That's the true k 7. End

From the foregoing, elbow method can be achieved by calculating the Sum Squared Error (SSE) which measures the sum of the squared distance between each data point and the centroid in the cluster it belongs. The optimal number of clusters is identified by the point on the graph that shows the elbow position. It can also be seen as the point where SSE starts to flatten. This point guarantees that when further K is added, there will not a change in the SSE. When the elbow method is run against the dataset, it suggest the number of optimum clusters needed for the clustering activity. This is referred to "K". K is the number of clusters and this is determined by selecting the point where the SSE starts to flattens. This is the amount of clusters that guarantees that when more cluster is added, there will be no change in the SSE values.

3.3.2 DBSCAN

DBSCAN(Density-Based Spatial Clustering of Applications with Noise) is a density-based clustering algorithm that identifies clusters based on the density of data points. "The density of a data point is defined by the number of points that lie within a radius Eps of that point (including the point itself). The densities of these spherical regions are used to classify the data points into core, border, or noise points." (Aggarwal 2015 p. 205). The algorithm dwells on three concepts; Core Points: the points with a minimum number of neighbouring points within a radius. Border Points: these are points that are withing the neighbourhood of a core point but have fewer neighbours than required. Noise Points: These are points that do not alight with any cluster.

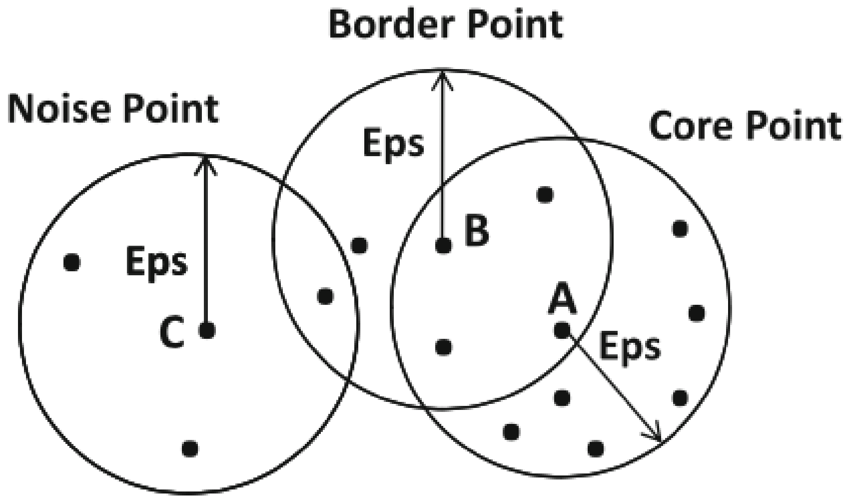


Figure 3.4: Aggarwal 2015 Fig. 6.16 Examples of core, border and noise points

According to Tan et al. (2019 p.349), the algorithm put into the same cluster any two core points that are close enough within a distance Eps of one another. In the same vein, any border point that is close enough to a core point is placed the same cluster as the core point. "Ties need to be resolved if a border point is close to core points from different clusters.) Noise points are discarded." (Tan et al 2019 p.349).

The algorithm operates on the following principles; 1. Point Classification: identify the points that are core, border and noise. 2. Noise Removal: get rid of the points are noise 3. Clustering Core Points: link all the core points within a certain distance of each other and put them in a cluster. 4. Border Point Assignment: assign each border point to the cluster of the core point closest to it. 5. Result: Each of the groups formed by the connected points form the clusters

"Because DBSCAN uses a density-based definition of a cluster, it is relatively resistant to noise and can handle clusters of arbitrary shapes and sizes. Thus, DBSCAN can find many clusters that could not be found using K-means" (Tan et al. 2019 p. 361)

3.4 Conclusions

In this chapter, we have seen the dataset used for the segmentation of customers. We have described the attributes and the licensing under which it is shared. The data processing and cleaning methods were outlined. These include handling missing values, normalization and category including. All these ensure that the dataset is prepared effectively and ready for the clustering algorithms. Two algorithms have been carefully selected as they can both handle a large dataset.

Chapter 4

Implementation

This chapter is a presentation of the application of K-means and DBSCAN algorithms on the customer dataset. The analysis done involves applying the two clustering methods on the dataset and showing varying visualization of the results and also identify the clusters to which each customer belongs.

4.1 Data Processing

The first step was to get the dataset ready. We started this by removing unwanted column and afterwards see what the fist 10 rows look like.

ID	Gender	Ever_Married	Age	Graduated	Profession	Work_Experience	Spending_Score	Family_Size
<int>	<chr>	<chr>	<int>	<chr>	<chr>	<int>	<chr>	<int>
462809	Male	No	22	No	Healthcare	1	Low	4
462643	Female	Yes	38	Yes	Engineer	NA	Average	3
466315	Female	Yes	67	Yes	Engineer	1	Low	1
461735	Male	Yes	67	Yes	Lawyer	0	High	2
462669	Female	Yes	40	Yes	Entertainment	NA	High	6
461319	Male	Yes	56	No	Artist	0	Average	2
460156	Male	No	32	Yes	Healthcare	1	Low	3
464347	Female	No	33	Yes	Healthcare	1	Low	3
465015	Female	Yes	61	Yes	Engineer	0	Low	3
465176	Female	Yes	55	Yes	Artist	1	Average	4

1–10 of 8,068 rows

Previous 1 2 3 4 5 6 ... 100 Next

Figure 4.1: Dataset View

Figure 4.1 shows there are 9 columns. The first column which is the "Customer ID" was removed as this has no relevance in the analysis.

It was observed that the data include several columns which empty values. This will definitely be a problem to the analysis. After cleaning the empty values, the data was reduced to 6718 from 8068.

```
#getting the data ready
```{r}
data <- read.csv("customer_data.csv", header=TRUE)
data <- data[,2:9]
data
```
```

Gender <chr>	Ever_Married <chr>	Age <int>	Graduated <chr>	Profession <chr>	Work_Experience <int>	Spending_Score <chr>	Family_Size <int>
Male	No	22	No	Healthcare	1	Low	4
Female	Yes	38	Yes	Engineer	NA	Average	3
Female	Yes	67	Yes	Engineer	1	Low	1
Male	Yes	67	Yes	Lawyer	0	High	2
Female	Yes	40	Yes	Entertainment	NA	High	6
Male	Yes	56	No	Artist	0	Average	2
Male	No	32	Yes	Healthcare	1	Low	3
Female	No	33	Yes	Healthcare	1	Low	3
Female	Yes	61	Yes	Engineer	0	Low	3
Female	Yes	55	Yes	Artist	1	Average	4

1-10 of 8,068 rows   Previous 1 2 3 4 5 6 ... 100 Next

Figure 4.2: Dataset view after cleaning

```
> summary(data)
       ID              Gender          Ever_Married            Age          Graduated          Profession        Work_Experience
 Min.   :458982   Length:8068        Length:8068        Min.   :18.00   Length:8068        Length:8068        Min.   : 0.000
 1st Qu.:461241   Class :character   Class :character   1st Qu.:30.00   Class :character   Class :character   1st Qu.: 0.000
 Median :463472   Mode  :character   Mode  :character   Median :40.00   Mode  :character   Mode  :character   Median : 1.000
 Mean   :463479                                         Mean   :43.47                                         Mean   : 2.642
 3rd Qu.:465744                                         3rd Qu.:53.00                                         3rd Qu.: 4.000
 Max.   :467974                                         Max.   :89.00                                         Max.   :14.000
                                                                                                              NA's   :829
 Spending_Score      Family_Size
 Length:8068        Min.   :1.00
 Class :character   1st Qu.:2.00
 Mode  :character   Median :3.00
                    Mean   :2.85
                    3rd Qu.:4.00
                    Max.   :9.00
                    NA's   :335
```

Figure 4.3: Summary of the dataset

The summary shows that some of the columns are characters and they need to be processed so they can be used for analysis. These columns are;

1. Gender 2. Ever_married 3. Graduated 4. Profession 5. Spending_Score

Profession: the column profession is a character that needs to be converted into a binary for ease of analysis. First we had to check the number of professions the dataset. The more profession the more difficult it will be to perform this task.

```
#checking for the number of profession in the dataset by checking unique values.
```{r}
unique(data$Profession)
```

 [1] "Healthcare"   "Engineer"   "Lawyer"   "Entertainment" "Artist"   "Executive"   "Doctor"   "Homemaker"
 [9] "Marketing"   ""
```

Figure 4.4: View of the profession

As seen in the figure 4.4, there are 9 unique values (profession) in the dataset. There is a 10th value which is an empty cell. To correct this, we converted every blank cell to “NA”. Clustering exercises are very difficult when there are empty values.

4.2 K-Means Clustering

The K-Means algorithm was applied to the dataset to group the existing customers in the dataset based on their similarities. The implementation stages will be discussed below.

Choosing the Number of Clusters (K) needed: as mentioned earlier, the elbow method was used to determine the optimal number of clusters. The method plots the sum of squared error against the number of clusters and selecting the point where the SSE starts to drop. This is the elbow point. The number taken from that point was used for clustering.

Determining and visualizing the optimal number of clusters

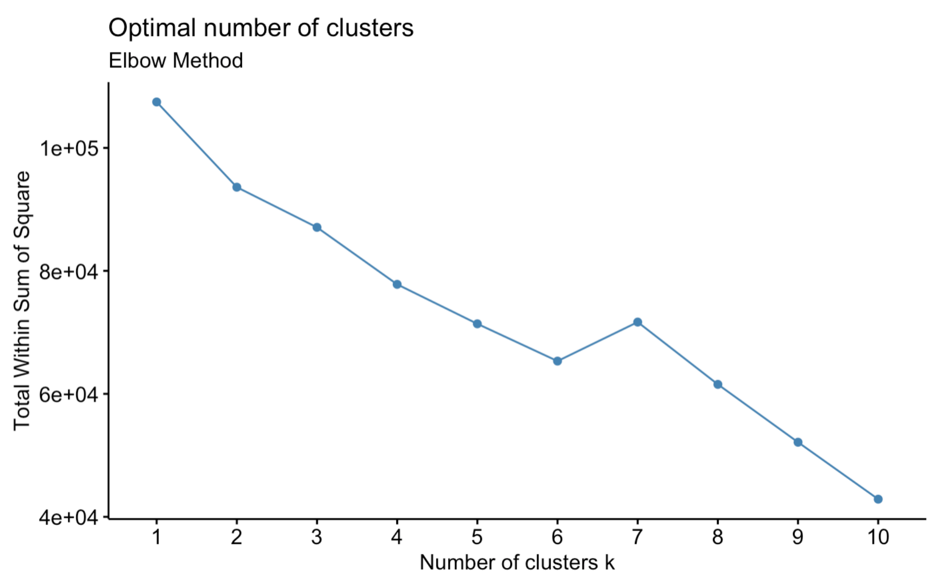


Figure 4.5: Elbow Method

Using the elbow method to determine the optimal number of clusters needed for the algorithm, we can see that the SSE starts to flatten at the 6th cluster. This is why we have used 6 as the number of clusters in the algorithm.

Silhouette Analysis

Before implementing the K-means clustering algorithm on the dataset for customer segmentation, the elbow method was used to determine the optimal number of clusters appropriate for the K-means clustering. This method is achieved by plotting the Within-Cluster Sum of Squares (WSS) against different values of K and identify the point where the graph flattens. This indicate the elbow point. Based on this method, we determined the optimal number of cluster result as “6”.

The next step is to validate the chosen number of clusters. To achieve this, we implemented the Silhouette Score into the K-means algorithm. The Silhouette score shows how well each points lies within its cluster compared to other clusters. This score normally ranges from -1 to 1. The value closer to 1 will indicate that the clusters are well defined while the score closer to 0 indicates there are cases of mis-clustering.

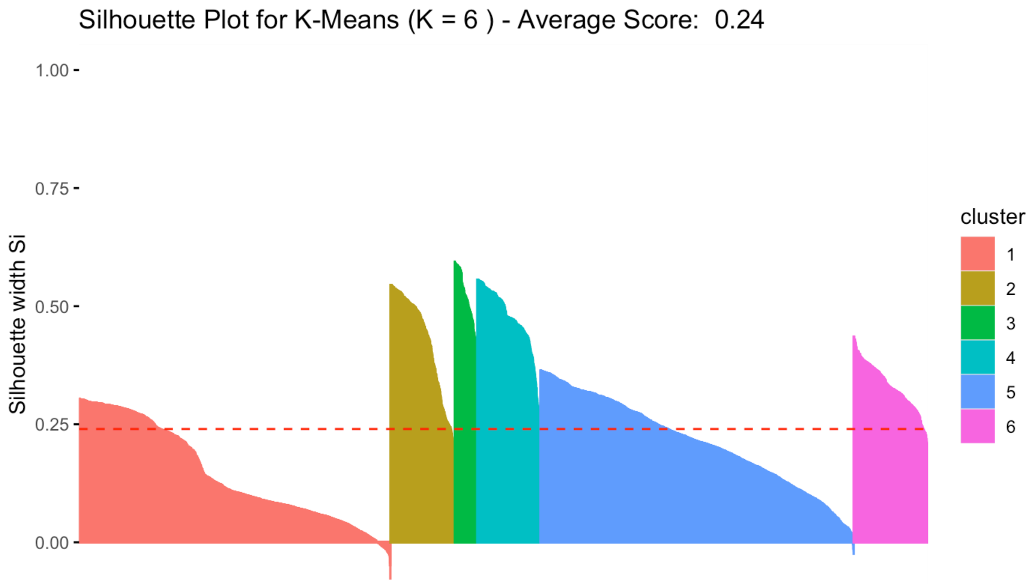


Figure 4.6: Silhouette Score

We computed the silhouette score and it came out as 0.239900165461564 which indicates a poor clustering. A score closer to 1 indicates good clustering while a score that is closer to 0 signifies an overlapping clusters.

Clusters

	cluster <fctr>	size <int>	ave.sil.width <dbl>
1	1	2464	0.14
2	2	509	0.42
3	3	178	0.52
4	4	501	0.47
5	5	2480	0.21
6	6	586	0.33

6 rows

Figure 4.7: Number of Clusters

Figure 4.7 shows the number of clusters and the size of the groups within the clusters.

	A	B	C	D	E	F	G	H	I
1	Clusters	Family_Size	Age	Work_Experience	Gender_Female	Ever_Married_No	Graduated_No	Doctor	Engineer
2	1	-0.272688	-0.3432359	0.2998129	0.459495155	0.538460593	-0.0846437	1.55301561	-0.309111676
3	2	0.63136475	-1.0378895	-0.024133436	-0.052720339	1.025660873	0.615467916	-0.311417709	-0.309111676
4	3	-0.1918473	-0.3010292	0.130316781	-0.33772187	0.641845289	0.015121587	-0.311417709	-0.309111676
5	4	0.08258836	-0.1147312	-0.0039977	0.709867392	-0.049530378	0.36603117	-0.311417709	3.234595213
6	5	-0.0520695	0.28223779	-0.076497772	-0.160303655	-0.830638414	-0.429066177	-0.30335346	-0.309111676
7	6	-0.1145683	1.20008597	-0.265876889	-0.202221261	-0.772463689	-0.04168569	-0.299209526	-0.309111676
8									
9									
10									
11									
12									
13									
14									

	J	K	L	M	N	O	P	Q
1	Entertainme	Executive	Healthcare	Homemaker	Lawyer	Marketing	Spending_Score_Average	Spending_Score_High
2	-0.3715439	-0.2828829	-0.4398109	0.81364789	-0.2810344	-0.1895355	-0.15514805	-0.332722944
3	-0.3715439	-0.2862963	2.22284128	-0.1649639	-0.280929	-0.1895355	-0.557373703	-0.3404186
4	1.16559745	-0.1963282	-0.4398109	-0.1649639	-0.2844665	1.01272171	-0.552530455	-0.311950981
5	-0.3715439	-0.2862963	-0.4398109	-0.1649639	-0.2844665	-0.1895355	0.128994997	-0.177816691
6	0.10879767	-0.1500309	-0.3916664	-0.1578362	-0.2692432	-0.1895355	1.098850753	-0.421106599
7	-0.3343891	1.08895301	-0.4304065	-0.1541736	1.34192187	-0.1800645	-0.558715865	1.647622369
8								
9								
10								
11								
12								
13								
14								

Figure 4.8: Exported cluster groups

	Profession_Healthcare	Profession_Homemaker	Profession_Lawyer	Profession_Marketing	Spending_Score_Average	Spending_Score_High	clusters$cluster
53	2.2733660	-0.1649639	-0.2844665	-0.1895355	-0.5767344	-0.4211066	6
53	-0.4398109	-0.1649639	-0.2844665	-0.1895355	-0.5767344	-0.4211066	1
53	-0.4398109	-0.1649639	3.5148294	-0.1895355	-0.5767344	2.3743421	4
53	-0.4398109	-0.1649639	-0.2844665	-0.1895355	1.7336423	-0.4211066	3
53	2.2733660	-0.1649639	-0.2844665	-0.1895355	-0.5767344	-0.4211066	6
53	2.2733660	-0.1649639	-0.2844665	-0.1895355	-0.5767344	-0.4211066	6
53	-0.4398109	-0.1649639	-0.2844665	-0.1895355	-0.5767344	-0.4211066	1
53	-0.4398109	-0.1649639	-0.2844665	-0.1895355	1.7336423	-0.4211066	3
53	-0.4398109	-0.1649639	-0.2844665	-0.1895355	-0.5767344	-0.4211066	1
53	2.2733660	-0.1649639	-0.2844665	-0.1895355	-0.5767344	-0.4211066	6
53	-0.4398109	-0.1649639	-0.2844665	-0.1895355	-0.5767344	-0.4211066	5
53	2.2733660	-0.1649639	-0.2844665	-0.1895355	-0.5767344	-0.4211066	6
53	-0.4398109	6.0610316	-0.2844665	-0.1895355	-0.5767344	-0.4211066	1
53	2.2733660	-0.1649639	-0.2844665	-0.1895355	-0.5767344	-0.4211066	6
53	-0.4398109	-0.1649639	-0.2844665	-0.1895355	1.7336423	-0.4211066	3
53	-0.4398109	-0.1649639	-0.2844665	-0.1895355	-0.5767344	2.3743421	5
53	-0.4398109	6.0610316	-0.2844665	-0.1895355	-0.5767344	-0.4211066	1
53	2.2733660	-0.1649639	-0.2844665	-0.1895355	-0.5767344	-0.4211066	6
53	-0.4398109	-0.1649639	-0.2844665	-0.1895355	-0.5767344	-0.4211066	5
53	-0.4398109	-0.1649639	-0.2844665	-0.1895355	-0.5767344	-0.4211066	5
53	-0.4398109	-0.1649639	-0.2844665	-0.1895355	1.7336423	-0.4211066	3
53	2.2733660	-0.1649639	-0.2844665	-0.1895355	-0.5767344	-0.4211066	6
53	-0.4398109	-0.1649639	-0.2844665	-0.1895355	1.7336423	-0.4211066	3
53	-0.4398109	-0.1649639	-0.2844665	-0.1895355	-0.5767344	-0.4211066	5
53	-0.4398109	-0.1649639	-0.2844665	-0.1895355	-0.5767344	-0.4211066	5
53	-0.4398109	-0.1649639	-0.2844665	-0.1895355	-0.5767344	-0.4211066	5
53	-0.4398109	-0.1649639	-0.2844665	-0.1895355	-0.5767344	-0.4211066	5
53	2.2733660	-0.1649639	-0.2844665	-0.1895355	-0.5767344	-0.4211066	6
53	-0.4398109	-0.1649639	3.5148294	-0.1895355	-0.5767344	2.3743421	4
53	2.2733660	-0.1649639	-0.2844665	-0.1895355	-0.5767344	-0.4211066	6
53	-0.4398109	-0.1649639	-0.2844665	-0.1895355	-0.5767344	-0.4211066	5
53	2.2733660	-0.1649639	-0.2844665	-0.1895355	1.7336423	-0.4211066	6

Figure 4.9: Matched Customers with clustering group

Figure 4.9 is an exported csv file that shows the original dataset bound with the clusters generated from the K-Means algorithm. Each of the customer in the dataset can be seen with the cluster they have been assigned.

Clusters	Family_Size	Age	Work_Experience	Gender_Female	Ever_Married_No	Graduated_No	Doctor	Engineer	Entertainment	Executive	Healthcare	Homemaker	Lawyer	Marketing	Spending_Score_Average	Spending_Score_High
1	(0.27)	(0.34)	0.30	0.46	0.54	(0.08)	1.55	(0.31)	(0.37)	(0.28)	(0.44)	0.81	(0.28)	(0.19)	(0.16)	(0.33)
2	0.63	(1.04)	(0.02)	(0.05)	1.03	0.62	(0.31)	(0.31)	(0.37)	(0.29)	2.22	(0.16)	(0.28)	(0.19)	(0.56)	(0.34)
3	(0.19)	(0.30)	0.13	(0.34)	0.64	0.02	(0.31)	(0.31)	1.17	(0.20)	(0.44)	(0.16)	(0.28)	1.01	(0.55)	(0.31)
4	0.08	(0.11)	(0.00)	0.71	(0.05)	0.37	(0.31)	3.23	(0.37)	(0.29)	(0.44)	(0.16)	(0.28)	(0.19)	0.13	(0.18)
5	(0.05)	0.28	(0.08)	(0.16)	(0.83)	(0.43)	(0.30)	(0.31)	0.11	(0.15)	(0.39)	(0.16)	(0.27)	(0.19)	1.10	(0.42)
6	(0.11)	1.20	(0.27)	(0.20)	(0.77)	(0.04)	(0.30)	(0.31)	(0.33)	1.09	(0.43)	(0.15)	1.34	(0.18)	(0.56)	1.65

Figure 4.10: Cluster Analysis

Figure 4.10 shows an attempt to analyse the clusters that have been identified by the K-Means clustering. I have used the random margin of two values; -0.35 and 0.35. any value higher than 0.35 will indicate an very strong positive.

Clusters	Label	Family_Size	Age	Work_Experience	Gender_Female	Ever_Married_No	Graduated_No	Doctor	Engineer	Entertainment	Executive	Healthcare	Homemaker	Lawyer	Marketing	Spending_Score_Average	Spending_Score_High
1	Doctors	(0.27)	(0.34)	0.30	0.46	0.54	(0.08)	1.55	(0.31)	(0.37)	(0.28)	(0.44)	0.81	(0.28)	(0.19)	(0.16)	(0.33)
2	Healthcare Parents	0.63	(1.04)	(0.02)	(0.05)	1.03	0.62	(0.31)	(0.31)	(0.37)	(0.29)	2.22	(0.16)	(0.28)	(0.19)	(0.56)	(0.34)
3	Married Entertainers	(0.19)	(0.30)	0.13	(0.34)	0.64	0.02	(0.31)	(0.31)	1.17	(0.20)	(0.44)	(0.16)	(0.28)	1.01	(0.55)	(0.31)
4	Male Engineers	0.08	(0.11)	(0.00)	0.71	(0.05)	0.37	(0.31)	3.23	(0.37)	(0.29)	(0.44)	(0.16)	(0.28)	(0.19)	0.13	(0.18)
5	Average Spenders	(0.05)	0.28	(0.08)	(0.16)	(0.83)	(0.43)	(0.30)	(0.31)	0.11	(0.15)	(0.39)	(0.16)	(0.27)	(0.19)	1.10	(0.42)
6	Old Lawyers/ExecutiBig Spenders	(0.11)	1.20	(0.27)	(0.20)	(0.77)	(0.04)	(0.30)	(0.31)	(0.33)	1.09	(0.43)	(0.15)	1.34	(0.18)	(0.56)	1.65

Figure 4.11: Matched Customers with labeled groups

Cluster 1: showed 1.55. for the variable Doctor. This is a very high figure. Other variables show a very low figure. Cluster 2: has a value of 2.22 for Healthcare. This indicates that these group work in the Health care sector. This groupp also shows a high value for family size and age. This could suggest that they are parents. Cluster 3: this group score very high in Entertainment. The score for Ever_Married also suggest that they are married. Cluster 4: the group score a top high score of 3.33 for Engineer. Cluster 5: has a high score of 1.10 for an average spending score. This suggests that these group of customer are the average spenders. Cluster 6: this group scores very height with a high spending score of 1.65, Executives at 1.09 and Lawyer at 1.34. with a high score of 1.20 in Age, its an indication that they are old.

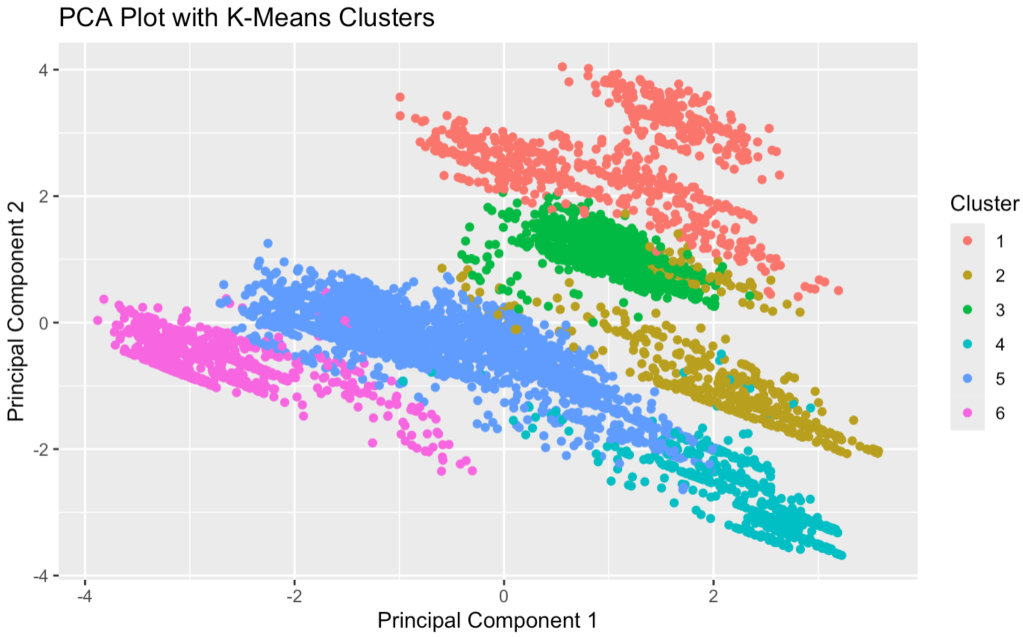


Figure 4.12: Visualization of clusters

4.3 DBSCAN Clustering

Selecting the Parameter: Epsilon (Eps) value defines the the radius of environment around a data point. It is also responsible for determining the minimum number of points required for a dense region (MinPts). The number of Eps and min sample parameters for the algorithm impacts the result of the clustering. For the purpose of this work, we have used 0.5 and 5 for Eps and MinPts respectively. The choice of these parameters are usually dependent on the characteristics of the dataset. These parameters were selected randomly and as a starting point in practical evaluations but can be fine-tuned based on the peculiarities of the given dataset. They also serve as a balance between identifying dense region and noise handling. The Eps parameters is the determinant of the distance between two points that made such point be considered as neighbours. When the value of Eps is small, it will result in a compact clusters. When the value is high, the clusters may become farther from each points. Hence, the reason for the 0.5. The MinPts parameters determines the minimum number of points that the algorithm needs to form a cluster. When the value of MinPts is high, the result will be fewer but well defined while a lower value will produce more clusters.

Silhouette Score:

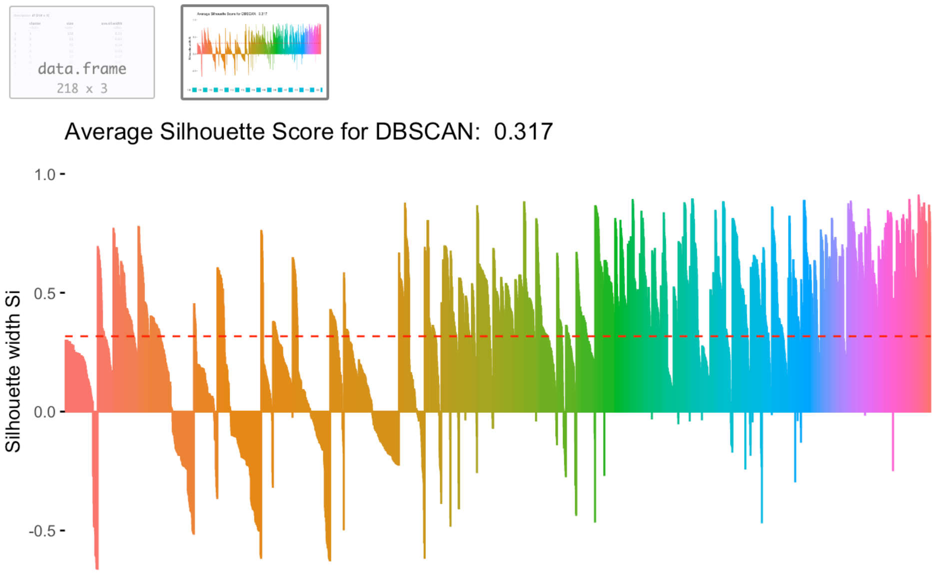


Figure 4.13: Visualization of the Silhouette

Visualizing the clusters

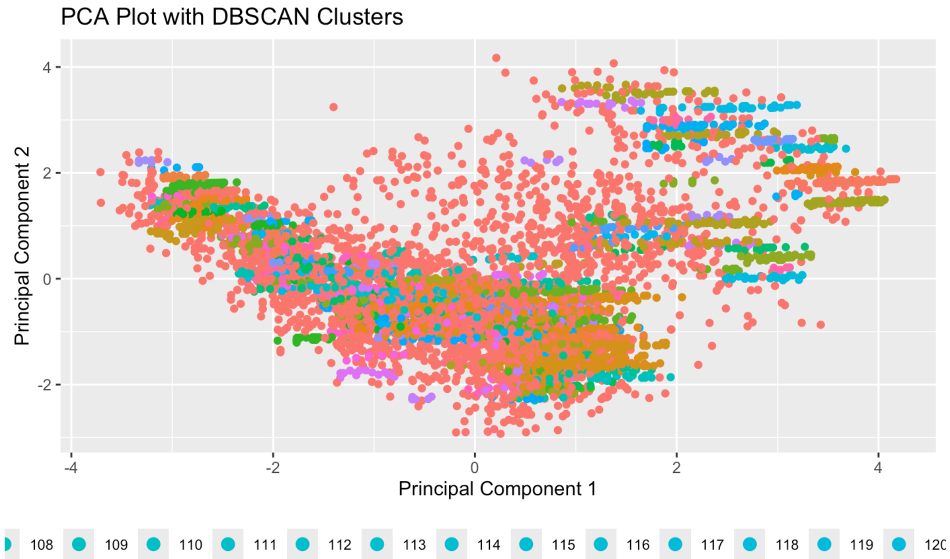


Figure 4.14: Visualization of clusters

Figure 4.14 is a visualization of the clustering group generated by the algorithm. Scatter plot was used to visualize the clusters formed by DBSCAN. Since the noise points are filtered by DBSCAN, the plot did not show the noise points. The noise points represent the customers who did not fit into any of the clusters.

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
2530	158	21	55	12	25	25	59	10	20	29	94	116	13	100	41	171	9	48	95	25	157	66	8	64	194
26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51
7	22	5	16	75	13	8	16	43	8	7	30	9	32	31	31	25	6	72	16	22	14	9	46	19	23
52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77
7	16	36	7	7	14	73	9	34	50	11	8	10	62	20	25	52	9	16	9	11	16	6	8	9	6
78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100	101	102	103
20	9	7	7	6	6	15	18	13	6	23	9	5	5	5	8	30	24	21	7	8	7	12	6	5	10
104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129
18	20	40	14	10	5	9	5	18	11	35	26	20	5	12	25	10	5	40	6	8	15	19	7	11	12
130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150	151	152	153	154	155
49	5	5	8	5	11	12	16	13	11	19	8	13	15	16	7	9	8	18	5	6	5	12	6	5	9
156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180	181
6	8	6	17	6	5	7	5	7	13	15	5	5	8	12	14	6	6	10	7	6	16	5	5	8	6
182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200	201	202	203	204	205	206	207
6	5	9	8	6	5	7	9	8	7	5	6	8	6	6	5	6	12	6	7	5	5	12	5	5	5
208	209	210	211	212	213	214	215	216	217	218															
5	6	5	5	4	5	5	5	5	5	5															

Figure 4.15: Visualization of clusters and data points

Figure 4.15 shows clearly the number of clusters and the number of data points that belong to each of the clusters. Cluster 0 is given to all the data points classified as noise. It shows a total 2,530 filtered as noise. These data points are not included the analysis.

4.4 Web Application Design

One of the objectives of this work is to design a simple application for users to perform clustering activities on a dataset. The purpose of this is to make clustering and its activities easy for anyone who do not have the background of data mining. These application is an attempt create a graphical user interface that is interactive and user friendly. Users can upload their dataset, process the data and apply the clustering algorithms. We have designed two; one for each of the clustering methods. The application provides visualization to help interpret the clusters.

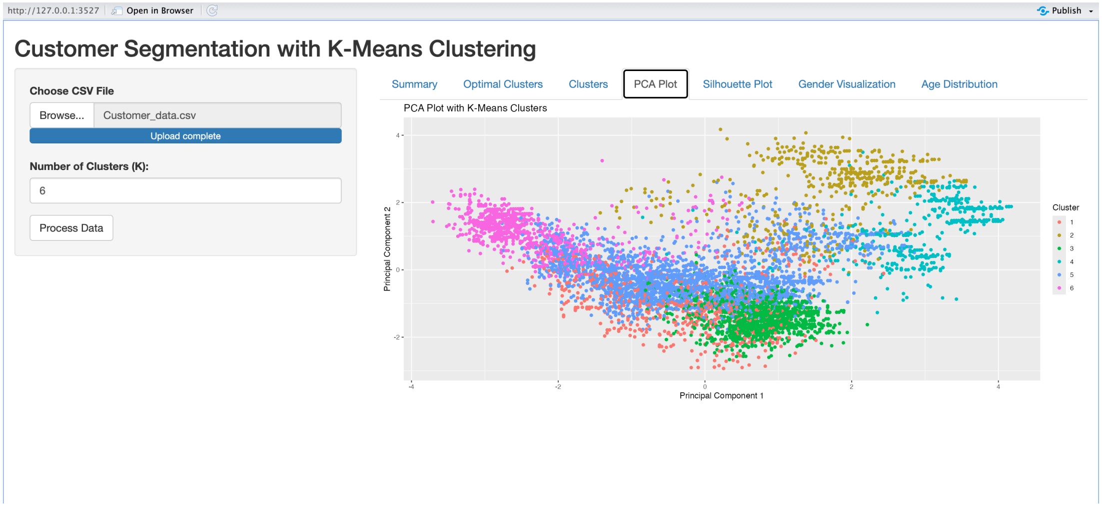


Figure 4.16: Web Application for K-Means

Figure 4.16 is the interface for K-Means clustering and it has the file upload function that allows users to upload data into the algorithm.

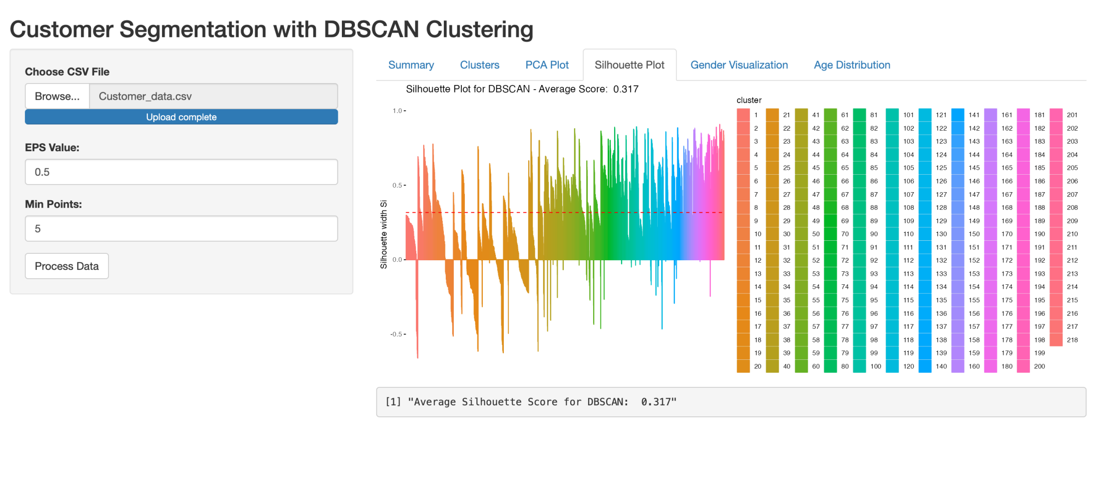


Figure 4.17: Web Application DBSCAN

Figure 4.17 also shows the interface for DBSCAN clustering. It has the file upload function with tuning parameter so select by the user. The application has the the following features; Data Upload: the application allows users to to upload their own CSV files. Immediately the data is uploaded, the data is processed automatically and ready for the clustering activities. Clustering: the application allows the clustering of data to be done. The algorithms that are integrated into it are K-Means and DBSCAN. Visualization: the application produces good visualization for the clustering and the silhouette. Plots: the application can also display different plots using the variables in the dataset showing different ranges.

4.5 Conclusions

The application of both K-means and DBSCAN provided a clear insights into the dataset used for the work. The reprocessing of data include data normalisation, scaling and principal component analysis for the reduction of dimensional. K-means presented a clear clustering of customers into 6 distinct groups based on their spending score and other demographics details. DBSCAN on the other hand presented 138 clusters and isolated 4,559 of the rows in the dataset as noise. This means these numbers are not clustered. The chapter also captured the visualization of the clustering for both algorithms. These help in understanding the strength and limitations of each algorithms. This allows for more informed decision making when it comes to customer segmentation and targeted advertisement.

Chapter 5

Evaluation & Testing

Introduction

In this chapter, the two clustering algorithms used (K-Means and DBSCAN) used for customer segmentation will be evaluated. The purpose of this evaluation and testing will be to compare the performance of each algorithm to the dataset. The two algorithms will be put to test based on various metrics such as their ability to discover meaningful clusters, handling of noise and outliers.

Evaluation Metrics

For proper evaluation of these two methods, the metrics used are listed below;

Clustering:K-means algorithm uses the Within-Cluster Sum of Square (WSS) to measure how tightly grouped the data points in the dataset are within each cluster. A lower WSS indicates a good clustering. This shows that the data points are closer to their respective cluster centroids.

Silhouette Score: This is the score used to measure how similar each of the objects is to its own cluster as compare to what it shares with other clusters. This is done to both K-means and DBSCAN to check how high the score is. Stability of Cluster:There are occasions where the algorithm will be run several times.

The stability of the clustering results at each of the conditions for K-means or the different Eps and MinPts values for DBSCAN will tell how they have both performed. Noise Handling:K-means algorithm do not make provision for Noise but DBSCAN does.

The ability to determine the noise points and clusters that are outliers is a very important feature to have. Dataset as the one used in this work mostly will have lots of outliers and should be taken into consideration.

Execution Time:These algorithm are usually tested on a large dataset. It is important to evaluate how efficient these algorithms are when computing these clusters. The efficiency will tell on the run time of the process.

Visualization: Given the nature of the dataset, most of the columns are not numerical, we had to convert the columns that are character into binary then scaled. The dataset couldn't be visualized so the Principal Component Analysis was used to reduce the dimentionality of the dataset. This made it possible to visualised the clustering results.

5.0.1 K-Means Clustering

Clustering: The elbow method was used to determine the optimal number of clusters needed to appropriately assign data points to clusters. The elbow method uses the WSS as a criteria for achieving this. When plotting the WSS against a range of cluster numbers, we chose the point where the graph starts to flatten.

Silhouette Analysis: the silhouette score was implemented on the K-means clustering and the result reads Average Silhouette Score for K = 6 : 0.239900165461564. an average Silhouette score of 0.24 indicates that the clustering is not very effective as this doesn't show a strong clustering. It shows that the data points are not well clustered. There is also a possibility that the points are not very distinct. It also implies that there is a chance that the clusters overlap. This happens when they are not well speared from one another. This will result in lack of clarity in the clustering results. Stability of Cluster:

To determine the stability of the clusters, we have run the clusters several times and there seems to be no change in the clustering down to the figures. Knowing well that the centroid selection for the clusters are random, this indicates a clear stability in the clustering.

Noise Handling: This is a limitation for K-means. It doesn't handle noise and outliers effectively. K-means assigns every point to a cluster. In cases where there are outliers, it will distort the centroids of the clusters. This will result to poor cluster assignments and ultimately affects the result of the clustering.

Execution Time: Considering the size of the dataset used, (6718 rows) K-means performed fairly. The time complexity is linear with respect to the number of data points and clusters in the dataset. K-means can be salable when used with large datasets. The execution time to find the optimal number of clusters (elbow) took almost two minutes (2Mins) but the clustering was quite fast.

Visualization: The Principal Component Analysis was used for the visualization and it uses two dimensions (2D). the plot showed a very clear cluster separations. This confirmed that K-means has successfully grouped customers that are similar in one cluster. It is also worthy of note that the PCA also shows slight overlaps in the clusters. This suggest that the algorithm couldn't successfully assign clusters to some of the data points.

5.0.2 DBSCAN Clustering

Selecting Parameter (Eps and MinPts): DBSCAN relies greatly on two parameters and they are Eps and MinPts. The Eps represent the maximum distance between two points to consider such data point a part of the same neighbourhood. The MinPts on the other hand is the minimum number of points required to form a dense region. Since the algorithm is a density based one, this points is very integral to its operation.

We experimented with different parameters and determined that

5 values for Eps = 0.5 and MinPts = 5 gave the best result for the dataset. This is also prevalent among the works previously done.

Silhouette Analysis: the silhouette analysis gave a score of 0.317 which was significantly higher than the K-means. DBSCAN seems better suited for a dataset with noise like ours.

Visualization: just like the K-means algorithm, we used the PCA for the reduction of dimensions. The DBSCAN clusters were visualized in a 2D plot. DBSCAN identified 138 clusters, and several data points as noise and outliers. With the DBSCAN algorithm, the clusters were well separated and the noise filtered out.

Noise Handling: DBSCAN stands out when it comes to the detection and treatment of noise in a dataset. The noise points were assigned a cluster label Zero (0). This made it easy for them to be excluded from the analysis. This feature of DBSCAN is very useful in identifying and managing outliers. These points could have distorted the results of the clustering.

Stability of Cluster: DBSCAN showed a very high cluster stability when it was rerun several times with the same parameters. Unlike K-means that relies on a random initialization, it ensures that same clusters are formed as long as the Eps and MinPts values do not change.

Execution Time: the DBSCAN execution time was a little longer than that of K-means. This could be attributed to the noise identification and filtering before it can assign cluster.

Both Algorithms at a glance

Metric	K-Means	DBSCAN
Silhouette Score	0.24	0.317
Parameters	Number of Clusters	Eps and MinPts
Visualization	Clear Separation	Overlapping
Stability of Cluster	Moderate and Depends on K	Highly Stable
Execution Time	Faster	Normal

Table 5.1: Comparing K-Means and DBSCAN algorithms

Web Application

The application, built with Shiny Package in R, could be said achieved more than 50% of the goal it was meant to serve. Having done most of the data processing, clustering and visualisation, the only task that is expected of the user is to upload the dataset and click on process data. The user do not know what happens in the back-end but is presented with the result of the clustering and visualization.

The application however, is not without limitations. It needs more flexibility, a more dynamic data handling. This will enable it accept any type of data.

5.1 Conclusions

It is clear that both K-means and DBSCAN have their respective strength and weakness. K-means performed well despite the size of the dataset which is large but we can conclude that it struggles with noise and outliers. All the data points in the dataset were assigned to a cluster so there is no way determine if all the assignments were correct. DBSCAN can be said to be a bit more reliable as it was able to identify the outliers labelling them all as noise. A total of 2530 points were labelled cluster 0 which is the label for data points that are not assigned. They are considered noise. This is almost half of the dataset. The total number of rows in the dataset is 6,718. Only 4,188 rows were assigned cluster.

K-means produced a compact and well separated clusters with a silhouette score of 0.24. DBSCAN with 0.217 indicates that it produced better defined clusters with their data points more cohesive and better separated. Given that DBSCAN produced a large number of noise filtering out lots of the dataset, it gives K-means an advantage.

We can conclude that DBSCAN performed better with a higher silhouette score. This indicates that the clusting quality are better in DBSCAN and it is not very strong for both K-means algorithms.

One core feature of DBSCAN is the ability to identify noise points. In the case of this dataset with 2530 noise points seems to be very large. This could suggest that the dataset has a complex structure that is not easily captured by DBSCAN. It also implies that the clusters are not well separated by density or that there are too many points that do not belong to any clear cluster.

In the same vein, we also considered that DBSCAN might be more suitable for datasets where noise filtering is more important than the cluster assignments. If the goal of the project is to cluster all the data, K-Means might be more appropriate since it will assign every data point to a cluster. However, there is a danger of forcing data points into clusters where they naturally do not belong.

Considering the performance, there has to be an evaluation of the specific goals of the clustering task. If there is need to cluster every data point then K-means is the algorithm and if the noise exclusion and retainining core clustering is the goal, DBSCAN does the job.

Given that DBSCAN produced a higher silhouette score (0.317 vs 0.24) and was able to successfully identify the noise points, it appears to be a better choice for the dataset. We cannot ignore the fact that DBSCAN labeled 2530 points as noise. This suggest that the dataset might contain a large number of outliers or points that do not fit well into any of the identified clusters.

The application that was designed was an attempt to automate customer segmentation thereby making the clustering and data mining process less cumbersome and attractive.

Chapter 6

Conclusion

The analysis so far have provided a clear insight into the clustering and segmentation of customers using a customer dataset. The implementation featured two popular clustering algorithm namely; K-Means and DBSCAN. The performance of each of the algorithm was looked into on the basis of quality of clusters, stability, execution time and their ability to handle noise and outliers.

6.1 K-Means

The K-means clustering method was implemented using six (6) clusters upon a verification from the elbow method analysis. Then the silhouette score was implemented to see the quality of the clusters. The score came back as 0.239 which is an indication of a weak clustering performance. This also suggests that there might be data points that are not separated and would mean overlapping. The K-Means was consistent as it returned the same result when run multiple times. However, it did not account for noise and outliers as it struggles to fit all the data points into a cluster. There is a chance that some of the data points do not belong in the group they were assigned. This in turn will cause doubts on the accuracy of the clustering. Visualization of the clusters was achieved through PCA and it did a good job revealing the distinct cluster groupings but definitely shows lots of overlapping. This reflected the problems and limitation of the cluster separation. Nonetheless, the K-Means algorithm showed some reasonable level of stability, scalability in terms of large dataset and the execution time was relatively good.

One clear disadvantage of K-Means clustering is it sensitive to the initialization of the centroids. Even where there is an obvious distance between the data points and the centroid, that point might just end up with such centroid. A poor initialization will definitely result in poor clustering. In future work, this can be overcome by using the K-Means++ instead. The K-Means++ ensures a smarter initialization of the centroids and it has been reported to show more stability and better quality of clustering. The K-Means++ isn't entirely a different algorithm but more like a hybrid because aside the initialization, every other parts are same with K-Means.

6.2 DBSCAN

One would be inclined to agree that the DBSCAN clustering was effective for this particular dataset because of its ability to detect and handle noise. The algorithm identified a total of 2,530 noise points. This is huge as it is almost half of the dataset but it gave the assurance that no data point was forced into a cluster group. With a silhouette score of 0.317, it performed better that K-Means which indicated a stronger clustering and better separation of clusters. DBSCAN identified 218 clusters and classified a lot of points as noise and they are excluded from the analysis. This is an advantage over K-Means. Identifying outliers and noise makes it more efficient for a dataset of this kind. The DBSCAN clustering also exhibited similar level of stability as it did not change the amount of clustering after several rerun with a fixed parameter. However, DBCAN produced a large number of clusters (218) which makes it impractical to manually assign customers to specific labels for targeted marketing like we did with K-Means. It will be very difficult to divide the possible group in the dataset along 218 clusters.

6.3 Observations

"Cluster analysis divides data into groups (clusters) that are meaningful, useful, or both. If meaningful groups are the goal, then the clusters should capture the natural structure of the data" Tan et al. p.307

This begs the question, which of the results churned by the two algorithms can be said to be most meaningful. ?

The two algorithm; K-Means and DBSCAN displayed their unique strengths and weakness, showing the importance of selecting the right clustering method that best suit the selected dataset. Every dataset has its characteristics, hence the reason to check the algorithm that best suits the dataset selected. This will guarantee a better result. K-Means offered a high level of simplicity and scalability but cannot handle noise and this will ultimately result in forced grouping and weakly separated clusters. DBSCAN on the other hand did very well identifying noise and wrongly shaped clusters. Though it produced too many clusters which do not give itself to customer labelling.

6.4 Future Work

In terms of further work, the analysis could be extended to automatically label and match each customer to a specific cluster in the DBSCAN results. This will go along way helping targeted marketing and it will improve the business ability to group customer based on their segmentation.

Preprocessing of the dataset posed a challenge to the visual representation of the certain relationship in the dataset. Spending score, for example, could have been compared relatively to gender and profession. This couldn't be achieved because the dataset had to be converted to binary and scaled during preprocessing. These visualizations could be enhanced giving better overall interpretations of the results if more sophisticated scaling techniques can be employed in the future.

As regard DBSCAN, Effort should be made to reduce the number of clusters to better organize as long as it will not compromise the quality of the clustering. With fewer clusters, the customers could be manageable for targeted advertisement. There could also be varying level of parameters until a suitable number of clusters and very high silhouette score is achieved.

DBSCAN performed better than K-means in the ascpect of ability to handle noise in the dataset as indicated previously, however, "DBSCAN has trouble when the clusters have widely varying densities. It also has trouble with high-dimensional data because density is more difficult to define for such data" TAN, P. et al 2019 p. 351).

Enhancement of Web Application

The web application was an attempt at demonstrating a practical automation of clustering algorithms but would definitely use some further development. It showed some level of practical implementation but has lots of limitations. One major improvement could be the integration of multiple clustering algorithms into the application to allow for real-time comparison of results. This will help businesses select the most appropriate method for their specific datasets.

Additionally, the application could be enhanced by integrating an export functionality. This will allow users export their clustering results for further analysis or reporting. This feature would definitely improve the usability and impact of the application in real world business settings. This way customer segmentation insights can be used for vital decision making in businesses.

In conclusion, this dissertation has successfully explored and implemented two clustering methods; K-Means and DBSCAN. It has explored the unique advantages they both have in customer segmentation. This results shows that the work has delivered on its aim which is to

to explore data analytics and mining techniques to improve segmentation of customers and planed targeted marketing strategies.

When business know their customer and can group them, it can help improve the advertisement targeting process and it saves the business money as they need not spend so much in targeting all their customers.

While K-Means proved to be stable and efficient, DBSCAN displayed a superior noise handling and clustering separation. This, however, comes at the cost of producing numerous clusters and lots of noise. Future work should focus on refining these algorithms for practical business application, especially in terms of segment labeling.

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